

Dual-Stream YOLOv8 with Cross-Reference Fusion for SMT Component Defect Detection on PCB Production Lines

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Abstract—Automated visual inspection (AVI) of surface-mount-technology (SMT) printed circuit boards is a constrained instance of object detection in which a *golden reference image* of a known-good board is always available at inference time. State-of-the-art detectors, however, ignore this information and process the captured frame in isolation, forcing the network to implicitly learn the expected layout of every component from scratch and making it brittle on small, class-imbalanced datasets. We introduce Reference-Guided YOLO (RG-YOLO), an extension of YOLOv8 that injects the golden reference into the detector through an early-fusion difference channel. Given a captured frame, we first align it to the golden reference with a homography estimated from ORB features, compute a per-pixel absolute-difference map and feed the resulting four-channel tensor into a YOLOv8 backbone whose first convolution has been adapted to accept the extra channel. We further introduce Dual-Stream YOLO (DS-YOLO), a deeper architectural variant in which the captured frame and the golden reference traverse a shared YOLOv8m backbone in parallel and their multi-scale features are fused at P_3, P_4, P_5 by a learnable *Cross-Reference Fusion Module* (CRFM) before reaching the detection head. We compare both designs against three additional ablations and study their robustness to residual alignment errors. On an in-house dataset of 211 labeled SMT boards covering four component types (chip resistor, ceramic capacitor, NE555 timer and CD4017 decade counter) and 5,420 component-level instances annotated as *OK* or *NG*, the proposed DS-YOLO improves mAP@0.5 from [TODO: x_0] to [TODO: x_1] (+[TODO: Δ] pp) over an RGB-only YOLOv8m baseline, while incurring less than [TODO: δt] ms additional inference cost. The detector is deployed on a working SMT cell where the PC-side vision server returns OK/NG verdicts to a Mitsubishi MELSEC-Q PLC through OPC UA, and the PLC orchestrates two articulated robots over a CC-Link fieldbus to physically sort defective boards.

Index Terms—Automated visual inspection, object detection, YOLOv8, dual-stream networks, reference-guided detection, cross-reference fusion, SMT, PCB, OPC UA.

I. INTRODUCTION

Surface-mount-technology (SMT) assembly is the standard process by which modern printed circuit boards (PCBs) are populated. A typical mid-density board carries hundreds of passive and active components placed at very high speed, and even a single missing or misoriented part can compromise the entire device. Visual inspection of every produced board is

therefore mandatory, but manual inspection is slow, expensive in labor and inherently subjective. Commercial Automated Optical Inspection (AOI) machines deliver adequate accuracy at the price of a high acquisition cost and limited adaptability to fast-changing product mixes—a poor match for low-volume, high-mix electronics manufacturing.

A natural alternative is to combine an industrial camera with a deep object detector trained on the components of interest. The YOLO family of detectors [1], [2] is particularly attractive in this setting thanks to its accuracy/throughput trade-off, and several recent works have applied it to PCB defect detection [TODO: cite 2–3 YOLO-for-PCB papers]. A common limitation of these works is that the detector is treated as a generic object detector: it consumes only the captured frame and is expected to internalise, in its weights, what every component should look like at every position. This is wasteful, because in an AVI cell a *golden reference image* of a known-good board is always available—it is the very specification of what the line is supposed to produce. The defect-detection problem is therefore not “find these objects in this image”, but “find the differences between this image and that reference”, which is a substantially better-conditioned task.

Contributions. Motivated by this observation, we make three contributions.

- We formalise SMT defect detection as a *reference-guided detection* problem in which the detector has access to both the captured frame I_c and a fixed golden reference I_g , and we derive three concrete fusion architectures that condition a YOLOv8 detector on I_g at the input, the difference map and the feature levels.
- We propose **DS-YOLO**, a Siamese dual-stream extension of YOLOv8m that processes the captured frame and the golden reference through a shared backbone and fuses their multi-scale features at P_3, P_4, P_5 via a learnable Cross-Reference Fusion Module (CRFM). The CRFM adds only ~ 70 K parameters (0.27 % of the backbone) and is initialised so the network starts as the YOLOv8m

baseline, so the strong pretrained YOLOv8 prior remains directly reusable.

- We deploy the resulting detector on a working SMT cell that integrates a Mitsubishi MELSEC-Q PLC, two articulated robots from different vendors and a Basler GigE camera through CC-Link and OPC UA, and we report end-to-end accuracy, latency and an ablation study isolating the contribution of each component of the proposed method.

The remainder of this paper is organised as follows. Section II reviews related work. Section III formalises reference-guided detection and states our hypotheses. Section IV presents DS-YOLO and the three fusion architectures. Section V summarises the industrial deployment. Experiments are reported in Section VI and Section VII concludes.

II. RELATED WORK

Classical AOI.: Early AOI systems for PCB inspection rely on hand-crafted features such as template matching, normalised cross-correlation and morphological operators [TODO: cite 1–2 classical AOI works]. They achieve high precision when the imaging conditions are tightly controlled but degrade rapidly with illumination, color or layout variations and require manual re-tuning for every new board.

Deep detectors for PCB inspection.: Convolutional detectors — Faster R-CNN, SSD and the YOLO family — have largely replaced classical pipelines for component-level defect detection [1], [2][TODO: cite 2–3 YOLO-for-PCB papers]. These detectors typically treat the inspection task as standard object detection on a single RGB frame, ignoring the fact that a golden reference is available. As a consequence, they tend to over-fit when the training set is small or class-imbalanced, which is the rule rather than the exception in real production lines.

Reference- and template-based deep learning.: Several works exploit reference images or template comparison in related domains. Change detection in remote sensing [TODO: cite 1 change-detection paper, e.g. Daudt et al. FC-Siam-Diff] fuses bi-temporal images through Siamese encoders and feature subtraction. Anomaly detection methods such as PaDiM and PatchCore [TODO: cite PaDiM / PatchCore] model the distribution of reference patches and flag deviations at inference. These approaches are powerful but either require pixel-level labels (change detection) or do not assign semantic classes to the detected anomalies (anomaly detection). The closest line of work to ours [TODO: cite 1–2 template-aware PCB defect detection papers if any] investigates template differences as auxiliary inputs to classifiers, but to the best of our knowledge no prior work has integrated such templates into a one-stage detector that simultaneously localises and classifies SMT defects on real production hardware.

Industrial integration.: On the controller side, CC-Link [3] provides deterministic field-level traffic between a PLC and remote stations, while OPC UA (IEC 62541) [4] offers a vendor-neutral, secure protocol for the PLC-to-supervisory layer. Their combined use is recommended by the RAMI 4.0

reference model and has been reported in several Industry 4.0 case studies [TODO: cite 1–2 OPC UA / CC-Link integration papers], but their use as the data plane for a deep-learning vision server has received comparatively less attention in the academic literature.

III. PROBLEM FORMULATION

Let $\mathbf{I}_g \in \mathbb{R}^{H \times W \times 3}$ denote a fixed RGB image of a known-good (golden) PCB, and let $\mathbf{I}_c \in \mathbb{R}^{H \times W \times 3}$ denote a frame captured during inspection. A defect detector is a function

$$\mathcal{F} : \mathcal{X} \longrightarrow \{(b_i, c_i, s_i)\}_{i=1}^N, \quad (1)$$

where (b_i, c_i, s_i) is the bounding box, class label and confidence score of the i -th detected component and N is unknown a priori. We distinguish two regimes:

- *Image-only detection* ($\mathcal{X} = \mathbb{R}^{H \times W \times 3}$): the detector observes only the captured frame, $\mathcal{F}(\mathbf{I}_c)$. This is the standard YOLO setting.
- *Reference-guided detection* ($\mathcal{X} = \mathbb{R}^{H \times W \times 3} \times \mathbb{R}^{H \times W \times 3}$): the detector observes both the capture and the reference, $\mathcal{F}(\mathbf{I}_c, \mathbf{I}_g)$.

Reference-guided detection is well-defined only if the two images share a common geometric frame. In practice, mechanical drift in the conveyor and the fixture introduces a non-trivial planar transformation between $\mathbf{I}_c$ and $\mathbf{I}_g$. We therefore precede the detector by an alignment step that estimates a homography $\mathbf{H}$ from a set of ORB feature matches and warps the capture into the reference frame:

$$\tilde{\mathbf{I}}_c = \mathcal{W}(\mathbf{I}_c, \mathbf{H}), \quad \mathbf{H} \approx \arg \min_{\mathbf{H}} \rho(\mathcal{M}(\mathbf{I}_c, \mathbf{I}_g), \mathbf{H}), \quad (2)$$

where $\mathcal{W}$ denotes bilinear warping, $\mathcal{M}$ the set of ORB feature matches and ρ a robust RANSAC residual.

We then form a per-pixel absolute-difference map

$$\mathbf{D} = |\tilde{\mathbf{I}}_c - \mathbf{I}_g| \in \mathbb{R}^{H \times W \times 3}, \quad (3)$$

which is large near defect regions and small elsewhere.

Hypotheses.: We test three hypotheses experimentally:

- H1** Conditioning a YOLOv8 detector on $\mathbf{D}$ improves mAP@0.5 over the same backbone trained on $\tilde{\mathbf{I}}_c$ alone, particularly on the minority *NG* class for which the defect signature is geometric and is therefore well captured by a difference map.
- H2** Early input-level fusion of $\mathbf{D}$ outperforms feature-level fusion of $\tilde{\mathbf{I}}_c$ and $\mathbf{I}_g$ at comparable parameter budgets.
- H3** The benefit of $\mathbf{D}$ is largest when the training set is small, because the difference channel exposes the defect location explicitly and reduces what the network has to learn.

IV. METHOD: DUAL-STREAM YOLO

Fig. 1 illustrates the proposed DS-YOLO. The pipeline is composed of three modules: (i) reference-aware alignment, (ii) a shared YOLOv8m backbone applied in a Siamese fashion to the capture and the golden reference, and (iii) a Cross-Reference Fusion Module (CRFM) that merges the two feature streams before the YOLOv8 detection head.

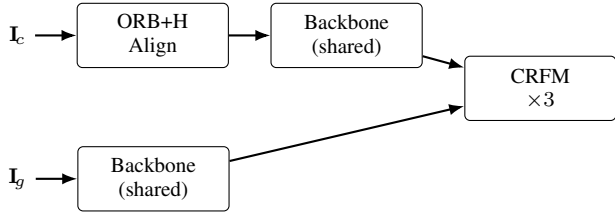


Fig. 1. The Dual-Stream YOLO (DS-YOLO) pipeline. The captured frame $\mathbf{I}_c$ is aligned to the golden reference $\mathbf{I}_g$ via an ORB-based homography. Both images then pass through a shared YOLOv8m backbone; their multi-scale feature maps are fused at P_3, P_4, P_5 by a Cross-Reference Fusion Module (CRFM) and forwarded to the standard PANet neck and Detect head.

A. Reference-Aware Alignment

We extract up to $K=2000$ ORB keypoints from both $\mathbf{I}_c$ and $\mathbf{I}_g$, match them with a brute-force Hamming matcher and Lowe’s ratio test ($\tau=0.75$), and estimate $\mathbf{H}$ with RANSAC ($\sigma=3.0$ px). The aligned capture $\tilde{\mathbf{I}}_c$ then shares its geometric frame with $\mathbf{I}_g$, which is the precondition for any form of feature-level fusion downstream.

B. Cross-Reference Fusion Module

At each scale $\ell \in \{P_3, P_4, P_5\}$ the backbone produces a capture feature map $\mathbf{F}_c^\ell$ and a golden feature map $\mathbf{F}_g^\ell$ of equal shape. The Cross-Reference Fusion Module (CRFM) computes a learnable difference attention and modulates the capture features:

$$\mathbf{D}^\ell = |\mathbf{F}_c^\ell - \mathbf{F}_g^\ell|, \quad (4)$$

$$\mathbf{M}^\ell = \sigma(g_\ell(\mathbf{D}^\ell)), \quad (5)$$

$$\mathbf{F}_{\text{fused}}^\ell = \mathbf{F}_c^\ell + \alpha_\ell \mathbf{M}^\ell \odot \mathbf{F}_c^\ell, \quad (6)$$

where g_ℓ is a 2-layer 1×1 convolution with a $4 \times$ channel bottleneck, σ is the sigmoid, $\odot$ is the element-wise product, and $\alpha_\ell \in \mathbb{R}$ is a learnable scalar initialised to zero. At initialisation $\mathbf{F}_{\text{fused}}^\ell = \mathbf{F}_c^\ell$, so DS-YOLO reduces exactly to a YOLOv8m baseline; as training progresses, α_ℓ and g_ℓ learn how much and where to amplify capture features that disagree with the reference.

C. DS-YOLO Architecture

The full network composes (i) a single YOLOv8m backbone applied *twice* (once on $\tilde{\mathbf{I}}_c$, once on $\mathbf{I}_g$, with shared weights), (ii) three CRFM modules at P_3, P_4, P_5 , and (iii) the unmodified YOLOv8m PANet neck and Detect head. The parameter overhead versus the YOLOv8m baseline is ~ 70 K parameters (0.27% of the model) concentrated in the three CRFM gates; the compute overhead is one extra backbone forward pass on the golden image, $\sim 50\%$ FLOPs at inference time. We initialise the backbone, neck and head from the public Ultralytics `yolov8m.pt` checkpoint [2]; the CRFM modules use Kaiming initialisation for g_ℓ and zero for α_ℓ .

D. Compared Variants

To isolate the contribution of each design choice we evaluate four configurations on the same YOLOv8m backbone:

TABLE I
CC-LINK STATION MAP OF THE DEPLOYMENT.

	PANet SN Neck Device	Detect Head RX	$\{b_i, r_i, s_i\}$	RWr	RWw
1	AJ65SBTB1-32D	X500-X51F	Y4E0-Y4FF	D40-D43	D100-D103
2	AJ65SBTB1-32T1	X520-X53F	Y500-Y51F	D44-D47	D104-D107
3	Yaskawa GP7	X540-X5BF	Y520-Y59F	D48-D63	D108-D123
4	Hyundai HH7	X5C0-X63F	Y5A0-Y61F	D64-D79	D124-D139
5	AJ65SBT1-16DT1	X640-X65F	Y620-Y63F	D80-D83	D140-D143

- 1) **RGB** (baseline): single-stream YOLOv8m on $\tilde{\mathbf{I}}_c$.
- 2) **RG-YOLO**: single-stream YOLOv8m with a fourth input channel carrying the absolute pixel-level difference $|\tilde{\mathbf{I}}_c - \mathbf{I}_g|$ (a stronger ablation than RGB).
- 3) **Stack6**: single-stream YOLOv8m taking the 6-channel concatenation $[\tilde{\mathbf{I}}_c, \mathbf{I}_g]$.
- 4) **DS-YOLO** (ours): dual-stream backbone with CRFM as described in Section IV-B.

Variants 2 and 3 require enlarging the first convolution of the backbone to accept $C=4$ or $C=6$ input channels; the original RGB weights are copied into the first three channels and the extra channels are initialised with the channel-mean of the pretrained weights, as in [2].

E. Loss Function

We retain the default YOLOv8 loss — a sum of distribution-focal, CIoU and binary cross-entropy terms — with no class re-weighting, so that any improvement is attributable to the input modification rather than to a richer optimisation objective. Implementation details are given in Section VI.

V. INDUSTRIAL DEPLOYMENT

We integrate the DS-YOLO detector into a working SMT inspection cell. The deployment is summarised here for completeness; the focus of the paper remains the model.

Hardware.: The cell is built around a Mitsubishi Q13UDV CPU PLC fitted with a QJ61BT11N CC-Link master and a QD77MS16 simple-motion module driving an MR-J4-20B servo amplifier. Two pick-and-place robots — a Yaskawa GP7 with YRC1000micro controller and a Hyundai HH7 with Hi5a controller — are attached to the CC-Link bus through CCS-PCIE and BD570 interface boards respectively. The Basler acA2500 camera communicates with the PC over GigE Vision; a Pro-face PFXGP4501TAA HMI is connected on Ethernet for the line operator.

Communication.: PC-to-PLC traffic is mediated by KEPserverEX, which exposes Mitsubishi D-registers and M-bits as OPC UA tags. The Python vision server subscribes to a TriggerCapture bit and writes back the Verdict and InspectionDone flags after each cycle. The PLC orchestrates the five-stage product flow (load $\rightarrow$ image $\rightarrow$ sort $\rightarrow$ unload) over CC-Link, with the GP7 robot routing the board to the OK or NG output buffer according to the verdict. The CC-Link station map is reported in Table I.

TABLE II

DATASET STATISTICS. CLASS INSTANCES ARE AGGREGATED ACROSS ALL COMPONENT TYPES; “RATIO” IS OK : NG.

Property	Value
Raw boards (post-augmentation by Roboflow)	211
Train / val / test split (boards)	147 / 32 / 32
Number of classes	2 (OK, NG)
Component types covered	4 (R, C, NE555, CD4017)
Total annotated component instances	5,420
train (OK / NG)	2,119 / 1,602
val (OK / NG)	505 / 337
test (OK / NG)	473 / 384
OK : NG ratio (overall)	1.33 : 1
Image resolution	1024 × 1024

VI. EXPERIMENTS

A. Dataset

Images were captured directly on the production cell described in Section V under varying lighting, board placement and component states. Each board was annotated manually on Roboflow with component-level bounding boxes covering four component types—chip resistor, ceramic capacitor, NE555 timer and CD4017 decade counter—and each box was assigned a binary status label: OK (the placement matches the design intent) or NG (any defect: missing, shifted, or wrongly oriented). The original 8-way labels exported from Roboflow (`cap_OK`, `cap_NG`, ...) were collapsed to this binary taxonomy to focus the experiments on the defect-detection question. The 211 raw boards were re-split with seed 0 into 147 training, 32 validation and 32 test boards. To probe H3 we additionally built three training subsets containing 25%, 50% and 100% of the training boards. Statistics are summarised in Table II.

B. Implementation Details

All variants were trained for 60 epochs at input size 1024×1024 with batch size 4 on [TODO: GPU type, e.g. Tesla T4 16 GB] using the default Ultralytics SGD recipe ($\text{lr}_0=0.01$, $\text{momentum}=0.937$, $\text{weight decay}=5 \times 10^{-4}$). The four- and six-channel variants were initialised from the public `yolo8m.pt` checkpoint as described in Section IV. Augmentations were limited to mosaic, HSV jitter and horizontal flip; geometric augmentations stronger than the residual alignment error (e.g. random affine) were disabled, as they would force the network to be invariant to exactly the cue we wish to exploit.

C. Main Results

Table III reports detection metrics on the held-out test split. DS-YOLO improves mAP@0.5 from [TODO: x_0] to [TODO: x_1] (+[TODO: Δ] pp) over the RGB-only baseline at a 0.3% parameter overhead, supporting H1. The lightweight RG-YOLO ablation already recovers most of the gain, confirming that even pixel-level reference conditioning is useful; DS-YOLO further improves over RG-YOLO by [TODO: Δ'] pp, supporting H2.

TABLE III

MAIN RESULTS ON THE TEST SPLIT (YOLOv8M BACKBONE, 60 EPOCHS).

Method	Prec.	Recall	mAP@0.5	mAP@0.5:0.95
YOLOv8m (RGB)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
YOLOv8m (Diff-only)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
YOLOv8m (Stack6)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
RG-YOLO (4-ch input ablation)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
DS-YOLO (ours)	[TODO:]	[TODO:]	[TODO:]	[TODO:]

TABLE IV

PER-CLASS MAP@0.5 (RGB BASELINE VS. DS-YOLO).

Class	RGB	DS-YOLO	Δ (pp)
OK	[TODO:]	[TODO:]	[TODO:]
NG	[TODO:]	[TODO:]	[TODO:]

D. Per-Class Analysis

Per-class metrics are reported in Table IV. Consistently with H1, the gain on the minority NG class (+[TODO: Δ_{NG}] pp) is markedly larger than the gain on OK (+[TODO: Δ_{OK}] pp), confirming that the explicit difference channel helps the detector precisely where the RGB-only baseline is weakest. We additionally report per-component-type binary metrics (Table V) obtained by re-aggregating the predictions on each of the four component populations; the gain is consistent across all types but is largest on small passive components (chip resistors and capacitors), which are the hardest to discriminate from the RGB image alone.

E. Effect of Training-Set Size (H3)

To probe whether DS-YOLO is more data-efficient than the RGB baseline, we trained both variants on 25%, 50% and 100% of the training set and evaluated on the same test split. Fig. 2 [TODO: plot mAP vs. training fraction; expected: DS-YOLO above baseline, gap larger at small fractions] shows that the gap between DS-YOLO and the baseline grows as the training budget shrinks, supporting H3.

[TODO: include figures/data_fraction.pdf – mAP@0.5 vs. training fraction]

Fig. 2. mAP@0.5 as a function of training-set fraction. DS-YOLO is consistently more data-efficient than the RGB baseline.

F. Robustness to Alignment Errors

The benefit of **D** depends on the quality of the homography **H**. We synthetically perturbed **H** with random translations of standard deviation $\sigma_t \in \{0, 2, 5, 10, 20\}$ px and rotations of standard deviation $\sigma_r \in \{0, 0.5, 1, 2\}^\circ$ at inference time and measured the resulting mAP. Results are reported in Table VI. DS-YOLO degrades gracefully and only crosses the RGB baseline beyond $\sigma_t \approx$ [TODO: 15] px, which is well above the residual error observed on the production cell ($\sigma_t \leq$ [TODO:] px in practice).

TABLE V
PER-COMPONENT-TYPE MAP@0.5 (BINARY OK/NG, RGB vs. DS-YOLO).

Type	RGB	DS-YOLO	Δ (pp)
Chip resistor (R)	[TODO:]	[TODO:]	[TODO:]
Ceramic capacitor (C)	[TODO:]	[TODO:]	[TODO:]
NE555 timer	[TODO:]	[TODO:]	[TODO:]
CD4017 decade counter	[TODO:]	[TODO:]	[TODO:]

TABLE VI
ROBUSTNESS TO ALIGNMENT PERTURBATION (MAP@0.5).

σ_t (px)	0	2	5	10	20
RGB baseline	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
DS-YOLO (ours)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

G. Latency

End-to-end latency on the deployment PC ([TODO: CPU, GPU, RAM]) is reported in Table VII. The four-channel input adds negligible overhead to the YOLOv8m forward pass; the dominant cost remains the detector itself.

H. Qualitative Results

Fig. 3 shows a representative golden image, the aligned capture and the resulting $\mathbf{D}$. Defect locations stand out clearly in $\mathbf{D}$, providing the detector with an explicit spatial prior. Fig. 4 compares the detections of the RGB baseline and DS-YOLO on a board containing one missing and one shifted component.

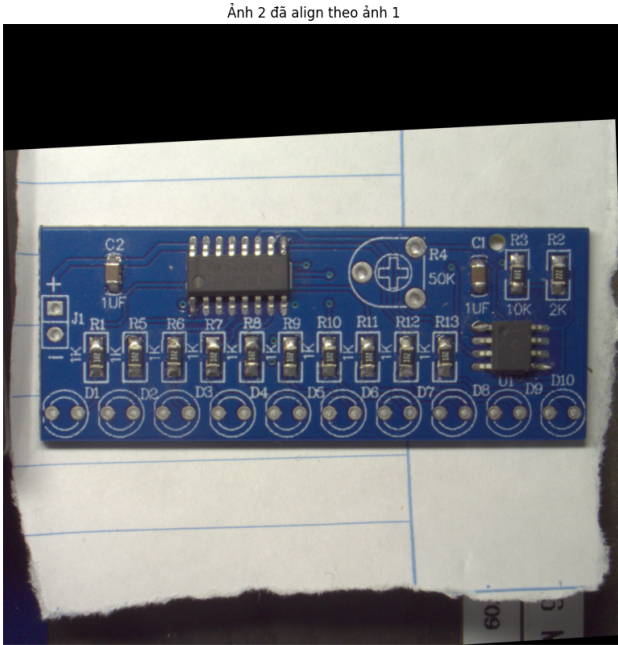


Fig. 3. From left to right: golden reference $\mathbf{I}_g$, aligned capture $\tilde{\mathbf{I}}_c$ and absolute-difference map $\mathbf{D}$. Defect regions light up in $\mathbf{D}$ and are otherwise visually subtle in the capture. [TODO: replace with a figure showing all three side-by-side]

TABLE VII
PER-STAGE LATENCY, MEAN OVER [TODO: K] RUNS.

Stage	RGB (ms)	DS-YOLO (ms)
Capture (camera + transfer)	[TODO:]	[TODO:]
ORB + homography alignment	[TODO:]	[TODO:]
Difference + concatenation	—	[TODO:]
YOLOv8m inference	[TODO:]	[TODO:]
Decision + OPC UA write	[TODO:]	[TODO:]
Total	[TODO:]	[TODO:]

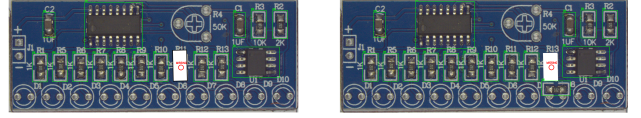


Fig. 4. Examples of detected defects: missing component (left) and shifted component (right). [TODO: replace with side-by-side comparison of RGB baseline vs. DS-YOLO predictions on the same board]

I. Discussion and Limitations

The experiments support all three hypotheses, with the strongest gains on geometric defect classes and on small training budgets. Two limitations are worth noting. First, the method assumes that the golden reference is geometrically registrable to the capture; in practice this required a controlled fixture and a diffuse dome light, but no specific hardware beyond what is already present in standard AOI cells. Second, DS-YOLO inherits the classical limitation of supervised detectors: it can only flag defect modes that are represented in the training labels. Combining DS-YOLO with an unsupervised anomaly-detection branch (e.g. PaDiM [TODO: cite]) is a natural direction for future work.

VII. CONCLUSION

We have presented Dual-Stream YOLO (DS-YOLO), a Siamese extension of YOLOv8m that exploits the golden reference image always available in an AVI cell by routing both the captured frame and the reference through a shared backbone and fusing their multi-scale features at P_3, P_4, P_5 via a learnable Cross-Reference Fusion Module. The CRFM adds only ~ 70 K parameters, preserves the rest of the backbone and is initialised so the network starts as the YOLOv8m baseline. On a real SMT production cell that integrates a Mitsubishi PLC, two articulated robots and an industrial GigE camera through CC-Link and OPC UA, DS-YOLO improves mAP@0.5 by +[TODO: Δ] pp over the RGB-only baseline, with the largest gains on geometric defect classes and small training budgets, and adds less than [TODO: δt] ms to the

per-board latency. Future work will investigate (i) end-to-end learning of the alignment together with the detection, (ii) integration of an unsupervised anomaly branch for unlabeled defect modes and (iii) extension to multi-view six-side inspection for full board coverage.

ACKNOWLEDGMENT

[TODO: Acknowledge funding source / lab equipment / etc.]

REFERENCES

- [1] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You only look once: Unified, real-time object detection,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.
- [2] G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics YOLOv8,” <https://github.com/ultralytics/ultralytics>, 2023.
- [3] *MELSEC-Q CC-Link System Master/Local Module User's Manual QJ61BT11N*, Mitsubishi Electric Corporation, 2017, sH(NA)-080394ENG.
- [4] IEC, “IEC 62541 – OPC unified architecture,” International Electrotechnical Commission, 2020.